Clustered Interval Calibration And Synthetic Segmentation Robustness

Uncertainty calibration across clustered data and segmentation stress cases

Research Presentation Program

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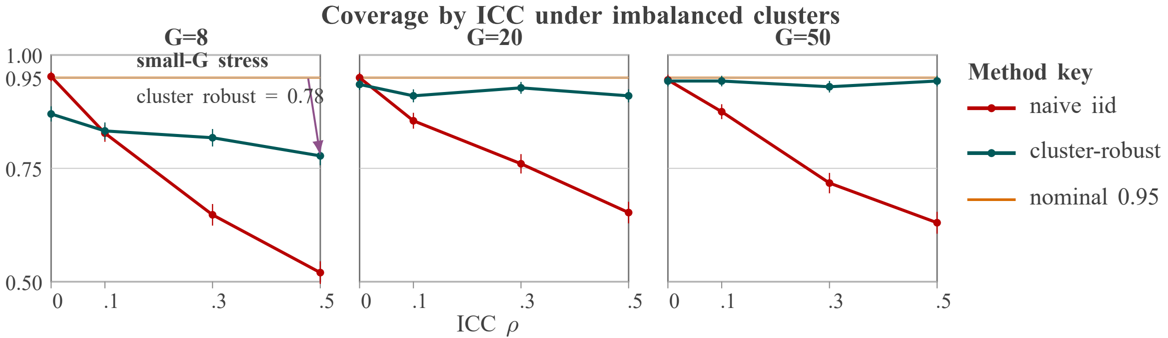
Clustered outcomes separate center and individual variation

$$Y_{ij} = \beta_0 + \beta_1 T_{ij} + u_j + \varepsilon_{ij}, \quad u_j \sim N(0, \tau^2), \quad \varepsilon_{ij} \sim N(0, \sigma^2)$$

Center effects set $\rho = \tau^2/(\tau^2 + \sigma^2)$, so interval calibration depends on cluster count.

Model components	**Interpretation**
• clustered outcome equation • center random effect • individual error	The source model makes center-level dependence explicit before interval comparison.

Cluster-robust intervals recover coverage as ICC increases



Data source: 400 simulation replicates per cell; y-axis shows 95% interval coverage.

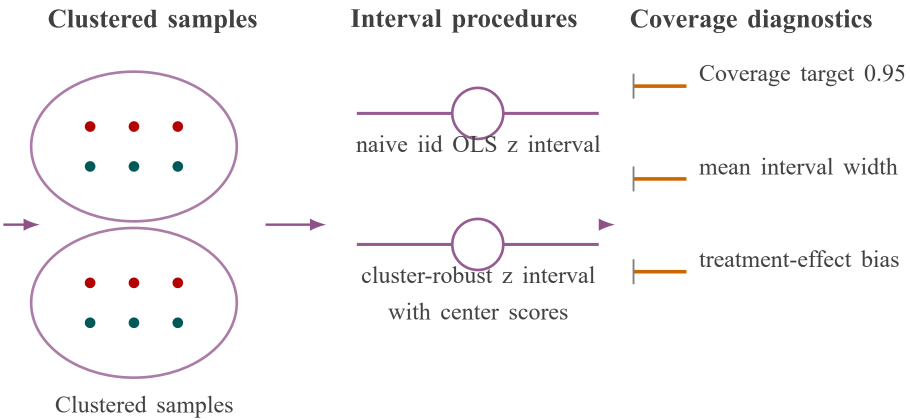
Small-G, high-ICC imbalance still suppresses coverage; center-robust intervals recover toward nominal as clusters increase.

Synthetic clustered-data example; intervals compared on coverage, width, and bias.

Simulation varies clustering before interval procedures are compared

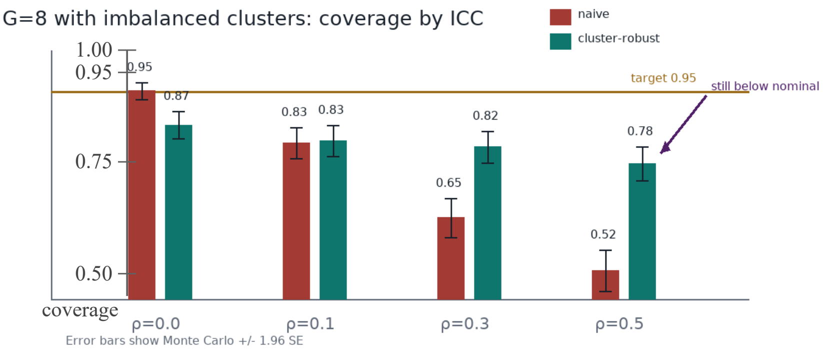
DGP knobs

- centers $G = 8, 20, 50$
- ICC $\rho = 0, .1, .3, .5$
- balanced or imbalanced clusters



The connector direction encodes data generation before interval estimation and coverage diagnostics.

Small-G settings remain anti-conservative after robustification



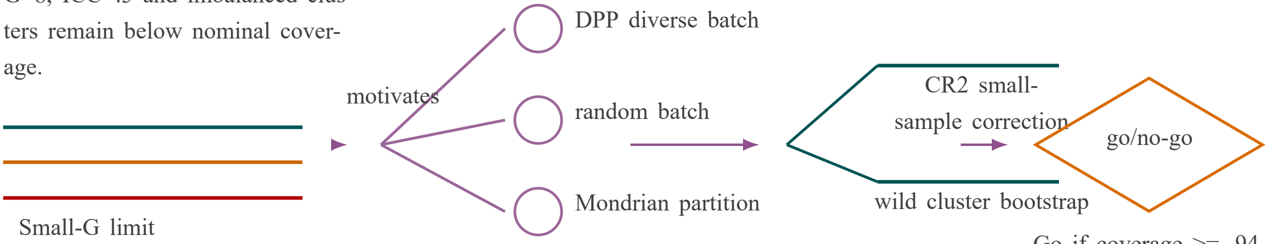
The failure region is adjacent to the diagnostic explanation instead of hidden in a note.

Negative evidence is completed; CR2 and wild cluster bootstrap remain the next test.

Next experiment tests whether batch selection reduces fragile coverage

Failure evidence | **Manipulate sampling** | **Comparator arms** | **Decision rule**

G=8, ICC=.5 and imbalanced clusters remain below nominal coverage.



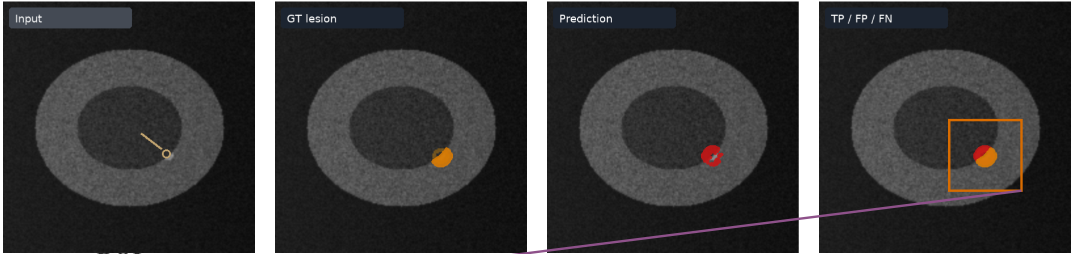
Go if coverage >= .94 with controlled width; no-go stratifies by Mondrian cell.

Coverage evidence determines which batch-query strategy to validate next; CR2 and wild bootstrap remain proposed comparators.

Same-case panels keep the segmentation error interpretable

Workstream transition **Segmentation robustness:** independent workstream; no causal bridge asserted.

Input **GT** **Prediction** **Error**



Same-case ROI zoom

GT crop Prediction crop Error crop

Overlay legend

- TP: overlap
- FP: prediction only
- FN: missed GT

All panels and zoom crops come from one case and one ROI; TP/FP/FN colors remain adjacent to the error crop.